

## Initial Project Planning Template

|               |                                                                  |
|---------------|------------------------------------------------------------------|
| Date          | 09-07-2026                                                       |
| Team ID       |                                                                  |
| Project Name  | Automated Credit Card Approval Prediction using Machine Learning |
| Maximum Marks | 5 Marks                                                          |

### Product Backlog, Sprint Schedule, and Estimation (4 Marks)

| Sprint  | Functional Requirement (Epic) | User Story Number | User Story / Task                                                                                               | Story Points | Priority | Team Members   | Sprint Start Date | Sprint End Date (Planned) |
|---------|-------------------------------|-------------------|-----------------------------------------------------------------------------------------------------------------|--------------|----------|----------------|-------------------|---------------------------|
| Sprint1 | Data Collection & EDA         | USN-1             | As a developer, I can collect the Credit Card Approval dataset and load it using Pandas.                        | 2            | High     | D Sarath Kumar | 01-072026         | 02-07-2026                |
| Sprint1 | Data Collection & EDA         | USN-2             | As a developer, I can perform exploratory data analysis to understand relationships between applicant features. | 3            | High     | V Raja         | 01-072026         | 02-07-2026                |
| Sprint1 | Data Preprocessing            | USN-3             | As a developer, I can handle missing values, encode categorical data, and scale numerical features.             | 2            | High     | D Sarath Kumar | 01-072026         | 02-07-2026                |
| Sprint1 | Data Preprocessing            | USN-4             | As a developer, I can split the dataset into training and testing sets.                                         | 1            | High     | V Raja         | 01-072026         | 02-07-2026                |
| Sprint2 | Model Building                | USN-5             | As a developer, I can train multiple Machine Learning models such as Logistic Regression, Decision Tree.        | 2            | Medium   | D Sarath Kumar | 02-072026         | 03-07-2026                |
| Sprint2 | Model Building                | USN-6             | As a developer, I can evaluate the models and select the best-performing algorithm based on accuracy            | 3            | High     | V Raja         | 02-072026         | 03-07-2026                |
| Sprint2 | Model Building                | USN-7             | As a developer, I can save the trained model and preprocessing objects using Pickle/Joblib.                     | 2            | High     | D Sarath Kumar | 02-072026         | 03-07-2026                |

|         |                         |        |                                                                                                                    |   |        |                |           |            |
|---------|-------------------------|--------|--------------------------------------------------------------------------------------------------------------------|---|--------|----------------|-----------|------------|
| Sprint2 | Model Building          | USN-8  | As a user, I can view the Home page describing the Credit Card Approval Prediction System.                         | 1 | High   | V Raja         | 02-072026 | 03-07-2026 |
| Sprint3 | Application Development | USN-9  | As a user, I can view the About page explaining the project and its objectives.                                    | 1 | Medium | D Sarath Kumar | 03-072026 | 04-07-2026 |
| Sprint3 | Application Development | USN-10 | As a user, I can enter applicant details such as age, income, employment status, credit history.                   | 1 | Low    | V Raja         | 03-072026 | 04-07-2026 |
| Sprint3 | Application Development | USN-11 | As a user, I can receive an instant prediction showing whether my credit card application is Approved or Rejected. | 3 | High   | D Sarath Kumar | 03-072026 | 04-07-2026 |
| Sprint3 | Application Development | USN-12 | As a developer, I can test the application using different applicant profiles                                      | 2 | High   | V Raja         | 03-072026 | 04-07-2026 |
| Sprint4 | Testing & Deployment    | USN-13 | As a developer, I can upload the project to GitHub for version control and collaboration.                          | 2 | High   | D Sarath Kumar | 04-072026 | 04-07-2026 |
| Sprint4 | Testing & Deployment    | USN-14 | As a developer, I can deploy the Flask application on Render so users can access it through a public URL.          | 1 | Medium | V Raja         | 04-072026 | 04-07-2026 |
| Sprint4 | Testing & Deployment    | USN-15 | As a developer, I can prepare the final project documentation and presentation for evaluation                      | 2 | High   | D Sarath Kumar | 04-072026 | 04-07-2026 |